Sensor-free shape estimation for a multi-section tendon-driven continuum robot

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1 Aim

Predict the whole backbone shape, tip position and tip orientation of a multi-section tendon-driven continuum robot in real time, from the tendon displacements alone (no sensors on the robot). A constant-curvature model gives a first estimate; a Gaussian-process model, trained on finite-element (SOFA) simulations of the CAD backbone, corrects it along the whole backbone. Motion capture on the physical robot is used for validation. Target: IEEE RA-L.

2 Status at a glance

Item	State	Status
Method: prior + learned correction along the backbone; tip-orientation model	Implemented and tested in simulation	**Done**
Simulation model: tendons	SOFA cables could push; now pull-only in every scene	**Fixed 29 Sep**
Simulation model: mesh	Mirror-symmetric mesh (20 k nodes per section), polar corotational elements: mirrored poses agree exactly	**Fixed 1 Oct**
Simulation model: gravity, material	Gravity was 1000 times too weak; TPU weight known, stiffness assumed	**Partly fixed**
Simulation model: disc contact	Modelled on the narrow-gap faces; used only for poses that close a gap	**Done 1 Oct**
Simulation speed	Large time step with early stop (about 13 times faster), contact only where needed	**Done 1 Oct**
Pretension	Straight two-section robot stands with 3 mm (falls with 2 mm); bent poses fold over at every value tested (3–12 mm)	**Needs stiffness**
Single-section sweep, realistic settings	76 poses done; tip error 14.7 → 0.16 mm with the learned correction	**Done 30 Sep**
Single-section sweep to 90° with contact	144 poses; running on the lab PC	**Running**
Two- and four-section sweeps	Gravity-free two-section sweep running; with gravity on hold until the stiffness is measured	**On hold**
Hardware validation (motion capture)	Not started; plan proposed in Section 5	**To plan**

3 Method

Figure 1 shows the pipeline. For a command (bend angle θ_j and direction α_j per section, converted to tendon displacements), SOFA gives the deformed body. The backbone centreline $\mathbf{c}(s)$ and cross-section orientation $\mathbf{R}(s)$ are extracted every 5 mm. The constant-curvature prediction is subtracted, and the remaining error curve is described by a few shape modes (functional PCA); one Gaussian

process per mode predicts its weight from the command. A separate Gaussian process predicts the tip orientation error. All results are held out by 5-fold cross-validation over poses.

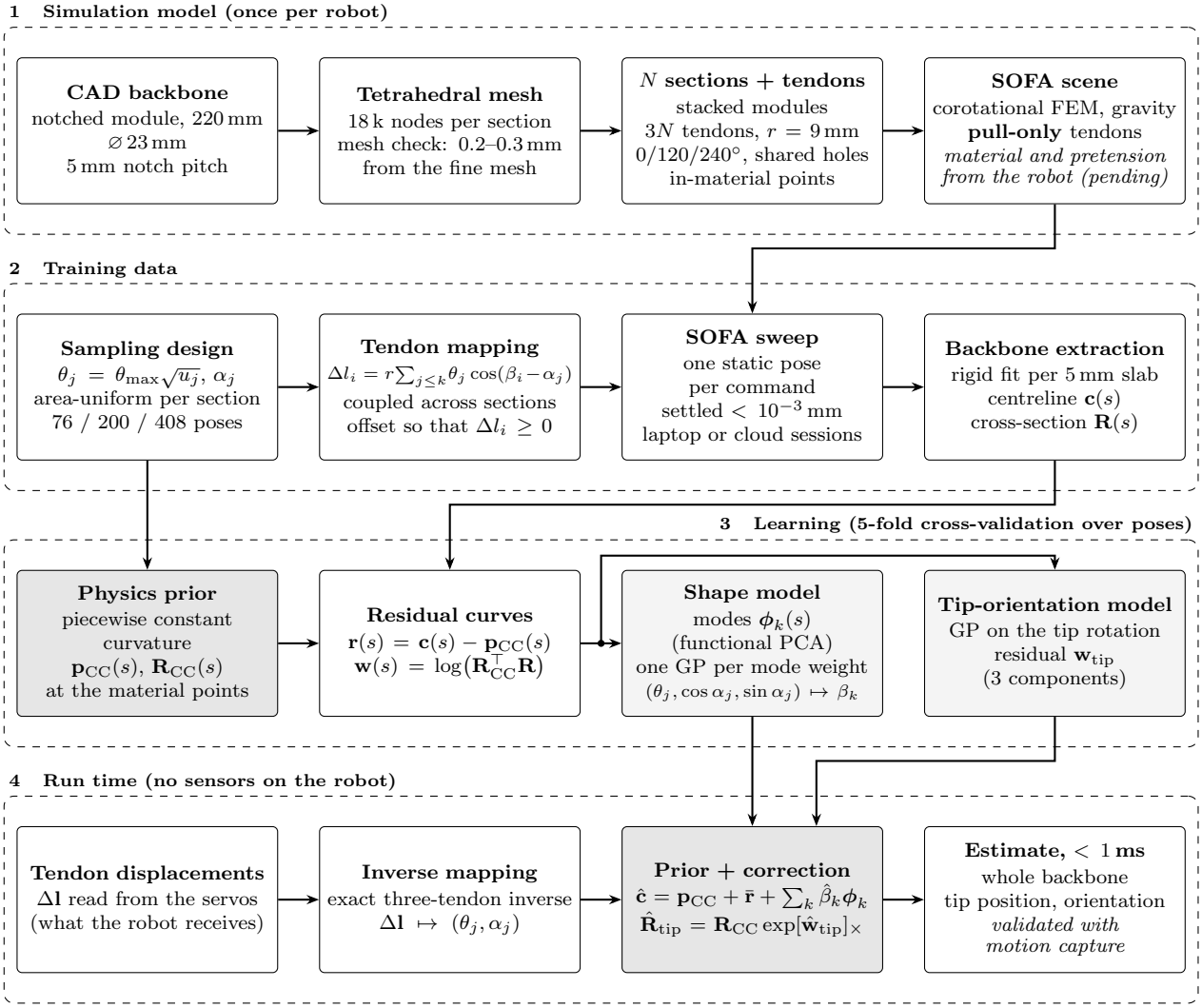


Figure 1: Pipeline: simulation model, training data, learning, and run time.

4 Results and findings

4.1 Single section with realistic settings (30 Sep)

CAD backbone, medium mesh, pull-only tendons, TPU weight (63 g), real gravity, stiffness 26 MPa (assumed). 76 poses, all settled, 5-fold cross-validation.

Held-out mean error	Tip (mm)	Along backbone (mm)	Worst point, 95% (mm)	Tip orient. (deg)
Constant curvature only	14.70	6.83	21.48	0.160
+ linear correction	1.63	0.88	2.69	–
+ Gaussian-process correction	**0.16**	**0.08**	**0.49**	**0.007**
Gaussian process without the physics prior	0.58	0.22	2.43	–

As a fraction of the 220 mm length: tip error 6.7 % with constant curvature alone, 0.07 % with the learned correction. Constant curvature already gives the tip orientation to 0.16°, because the tendon lengths fix the total bend; its position error comes from gravity moving the bend toward the base and from axial shortening. One shape mode carries 92 % of the error (Fig. 2). These are simulation-to-simulation numbers: they show what the method learns, not yet how well SOFA matches the robot.

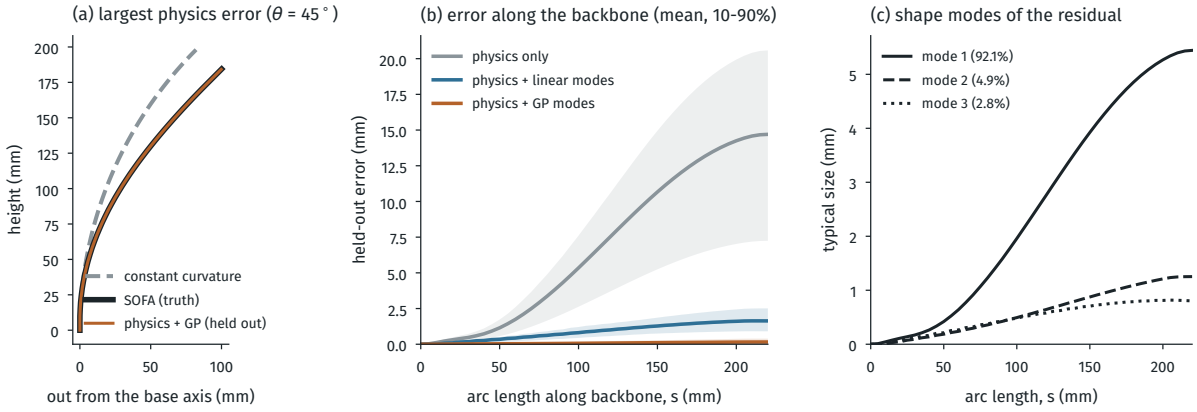


Figure 2: Single section, realistic settings. (a) Pose with the largest constant-curvature error. (b) Held-out error along the backbone. (c) The three largest shape modes of the error.

4.2 Learning results on the earlier, idealised simulation

These show that the method works. They were obtained before the corrections of Section 4.3 (tendons able to push, almost no gravity), so the numbers will change when the sweeps are repeated.

Held-out mean error	Tip (mm)	Along backbone (mm)	Tip orientation (deg)	Poses
One section, constant curvature only	3.24	1.66	0.76	76
One section, with learned correction	0.11	0.06	0.03	76
Two sections (CAD), constant curvature only	7.24	4.19	0.19	200
Two sections (CAD), with learned correction	0.79	0.30	0.06	200

A linear correction is clearly worse than the Gaussian process (two sections: tip 2.78 mm), and a Gaussian process without the physics prior is worse still (3.86 mm).

4.3 Corrections to the simulation model

(a) Tendons could push (found 29 Sep). A SOFA cable without a minimum force holds its length in both directions. In the old data, 173 of 200 two-section poses and 403 of 408 four-section poses relied on at least one pushing tendon. All scenes now use pull-only cables. Effect, two-section CAD model, same commands:

Command	Section 1 bend	Section 2 bend	Tip shift vs pushing
Section 2 only, 45°: pushing tendons	0.1°	44.9°	–
Section 2 only, 45°: pull-only	22.5°	22.5°	85 mm
S-shape 45°/45°: pushing tendons	44.9°	45.0° (back)	–
S-shape 45°/45°: pull-only	44.9°	0.0°	88 mm

Reason: section-2 tendons run through section 1, so a tendon fixes the *total* bend along its path. The geometric tendon mapping sets lengths but does not command the opposing tension that would keep section 1 straight.

(b) Mesh (30 Sep). Three single-section poses compared with a fine mesh (31.9 k nodes): the old mesh (11.5 k nodes) differs by 0.7–1.1 mm at the tip, the medium mesh (18.0 k nodes) by 0.15–0.29 mm. The medium mesh is now used.

(c) Gravity and material (30 Sep). The scenes use mm and kg, so gravity should be 9810 mm/s^2; it was 9.81. The backbone is TPU 95A at 100 % infill: 63 g per section (CAD volume 52.6 cm^3). Stiffness is set to 26 MPa, a typical datasheet value, not yet measured on the printed part. With these values:

- One section, 45°: the total bend stays 44.9° and the tip orientation is within 0.1° of constant curvature, but the tip position is 22.7 mm (10 % of length) from constant curvature; gravity moves the bend toward the base.
- Two sections, no pretension: the robot falls over even at zero command (section 1 bends 137°). The tendons hold their lengths; the backbone compresses by 4.9 %. The physical robot also needs pretension to stand, so pretension must be modelled.
- Two sections with pretension on every tendon, zero command (test of 30 Sep): at 26 MPa the robot falls with 0.5 mm and stands with 4 mm (1 and 2 mm still running); at 50 MPa it falls with 0 mm and stands with 1 mm. Pretension makes the simulated robot stand, as on the real one; the amount depends on the stiffness.

(d) Two sections under gravity, bent poses (1 Oct). Mirror-symmetric mesh, 26 MPa, 63 g per section, no contact:

- Zero command: the robot falls with 2 mm of pretension and stands with 3, 4, 6, 8 and 10 mm (tip within 2.5–5.4 mm of the axis).
- Bent commands (both sections 45° in one direction; section 2 only; S-shape; a random pose): the robot folds over with 3, 4, 6, 8 and 12 mm; the tip ends below the base. Pretension cannot prevent this: the backbone folds about the taut tendons by closing the notches on the inner side, which only disc contact stops.
- Without gravity the same poses settle normally (about 25 steps).

Either the printed backbone is stiffer than 26 MPa, or the real robot also folds onto its discs in bent poses. One stiffness measurement on a single section decides which, and it determines how the multi-section data are generated.

4.4 Computing

A lab PC (i9-14900K, 64 GB) is set up for the sweeps and reproduces the laptop's result for a check pose to 0.0004 mm. Sweeps can also run in cloud sessions. All code, meshes and results are in a private GitHub repository with continuous commits.

Speed (1 Oct). A time step of 1 s with early stopping reaches the same equilibrium as 400–800 small steps. With contact detection a two-section step costs 12.7 s instead of 1.9 s, so each pose is first settled without contact and repeated with contact only if a notch gap ends below 0.08 mm. Without friction an open gap carries no force: six test poses whose gaps stayed open gave identical results (0.0000 mm) in 27–44 % of the time, and the seventh closed a gap (0.03 mm) and is the kind that is repeated with contact. Poses in contact can alternate between two contact states every step (1.3 μm apart), so a settling check over an odd number of steps never passed; over 6 steps a 60° test pose stops at step 60 instead of the 400-step limit.

5 Decisions needed and proposed plan

1. **Paper story.** With pull-only tendons the sections are strongly coupled, and the geometric mapping misses the commanded pose by about 85 mm on two sections. Proposed: learn the real behaviour (tendon displacements to shape), and use the same model to correct the command, rather than only learning the constant-curvature error.
2. **Pretension.** On the robot it is "just enough to keep the backbone straight". In simulation this is 3 mm at 26 MPa (two sections). Bent poses are a separate issue (Section 4.3 (d)) that pretension does not solve.
3. **Measurements on the robot** (about 10 minutes): stiffness of one section (tip deflection under a known weight, or the sag of the section held horizontally); weight of one section; whether the two-section robot holds both sections at 45° without folding onto its discs; a side photo of "section 2 only, 45°" to check the coupling.
4. **Real data.** Goal: a real-time digital twin trained on SOFA, validated with motion capture. Proposed collection for two sections: about 70 poses (40 to calibrate the simulation-to-robot gap, 20 held out, 10 repeated from different directions for hysteresis). Rule of thumb for calibration: about 10 poses per input, i.e. about 20, 40 and 80 poses for one, two and four sections.
5. **Known gaps in the simulation:** tendon friction (steel line on TPU), sections held together only by tendon tension (modelled as joined), backbone stiffness (assumed, not measured).

6 Log

28 Sep Project moved to a private GitHub repository. Backbone-shape learning implemented for any number of sections. Two-section CAD sweep (200 poses) completed on the laptop. Area-uniform sampling design introduced.

29 Sep Tip-orientation model added. Pushing tendons found and fixed (pull-only). Four-section cloud sweep stopped because it used pushing tendons. Literature on coupling between sections collected (`docs/literature.md`, Section 6).

30 Sep Mesh check; medium mesh adopted. Gravity and units corrected; TPU weight set. Overnight laptop run failed (out of memory); sweeps moved to the lab PC. Single-section sweep with realistic settings completed (tip error 14.7 → 0.16 mm, orientation 0.16° → 0.007°). Two-section pretension test: the robot stands with a few millimetres of pretension. Closest related work identified (Kasaei et al.).

1 Oct Mesh made mirror-symmetric with polar elements (mirrored poses exact, so every pose also gives its mirror image). Large time step with early stopping (about 13 times faster, same equilibrium). Disc contact added on the narrow-gap faces; single-section sweep extended to 90°. Shared work queue and live progress display on the lab PC. Two-section sweep was too slow; causes found: contact detection (12.7 vs 1.9 s per step), poses in contact alternating between two states, and the robot folding over under gravity in bent poses. Contact is now used only where a gap closes, with identical results. Gravity-free two-section sweep started; the sweep with gravity waits for a stiffness measurement.